Biochemistry Cumulative Exam

(Supplemental Information Attached)

A mouse is fed a diet including the amino acid alanine, but no carbohydrates, or fatty acids. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

1. (40) Outline a metabolic pathway for the mouse to synthesize glutamate on this diet using alanine as the sole source of carbon, nitrogen and energy. How many alanine molecules must be used to provide enough carbon atoms, nitrogen, ATP, and NADH molecules to synthesize each molecule of glutamate? Write a complete, balanced equation for this process giving the numbers of ATP, NADH, $FADH_2$ and CO_2 generated.

2. (40) Outline a metabolic pathway for the mouse to synthesize aspartate on this diet using alanine as the sole source of carbon, and energy. How many alanine molecules must be used to provide enough carbon atoms, ATP, and NADH molecules to synthesize each molecule of aspartate? Write a complete, balanced equation for this process giving the numbers of ATP, NADH, $FADH_2$ and CO_2 generated.

3. (20) If alanine is labeled with ^{14}C at carbon 3, which carbon(s) of glutamate and aspartate will first become labeled?

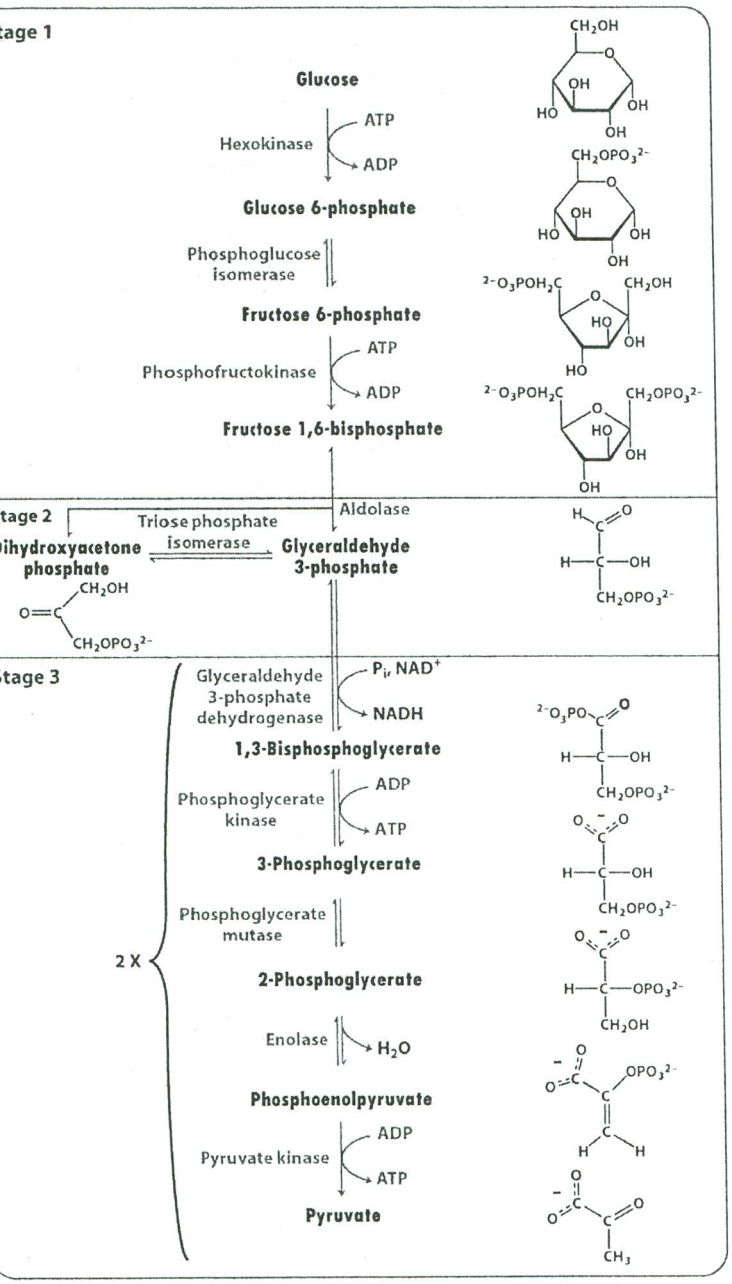

Figure 16-2
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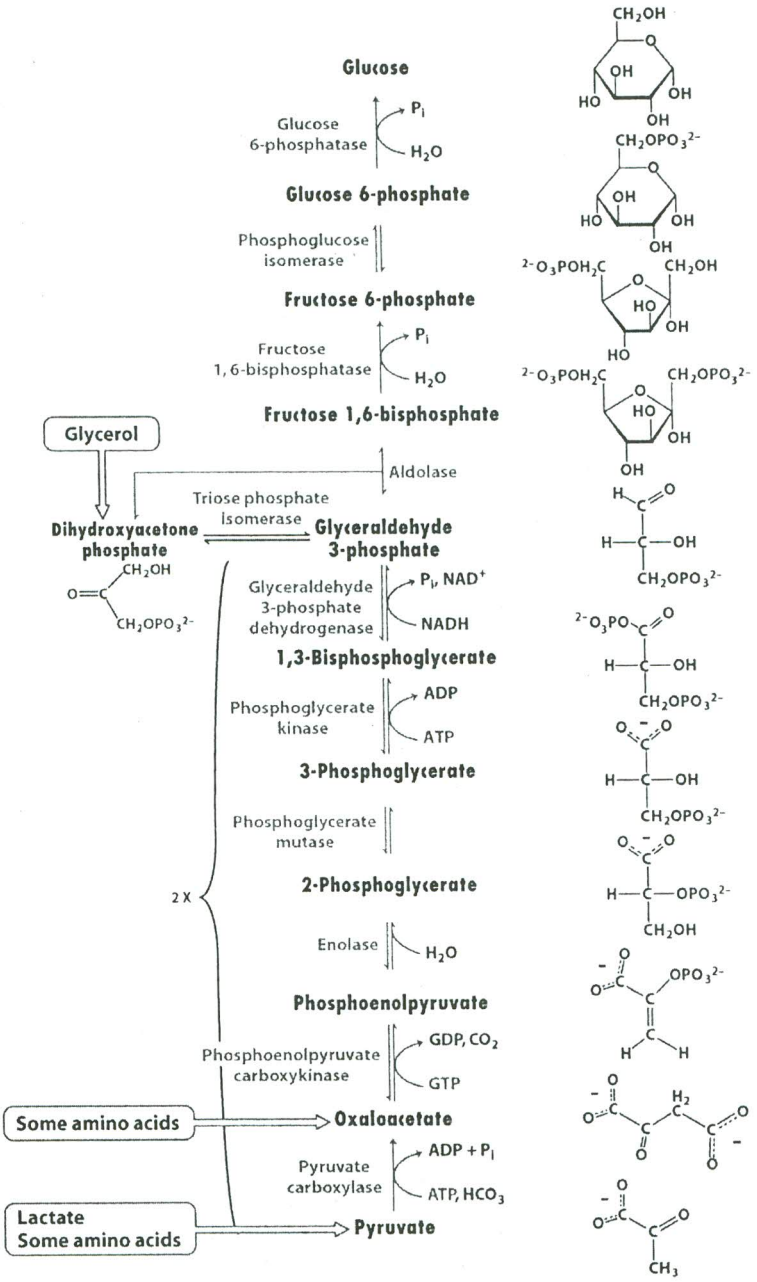

Figure 16-22
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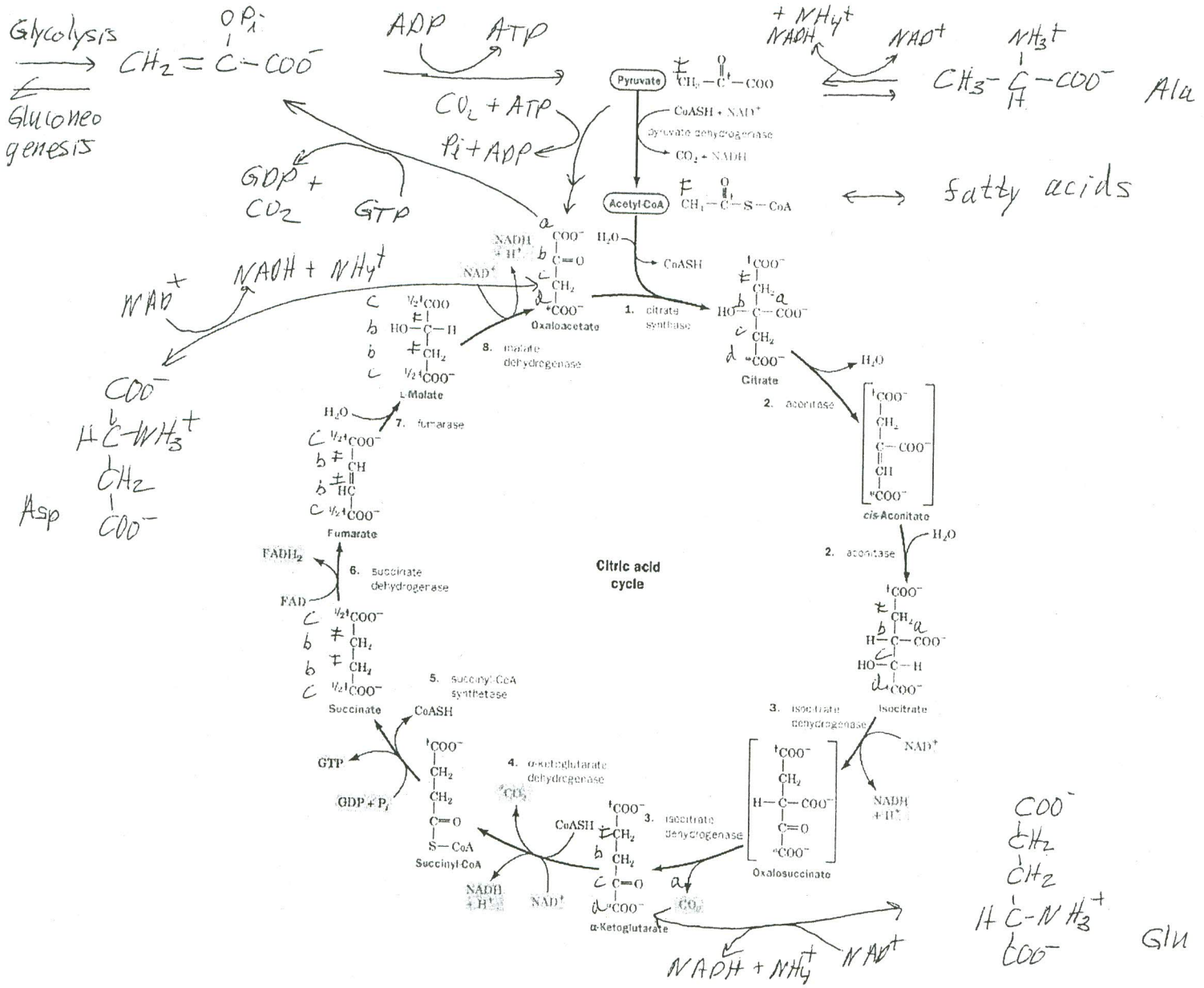
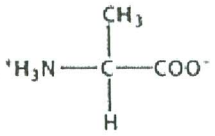
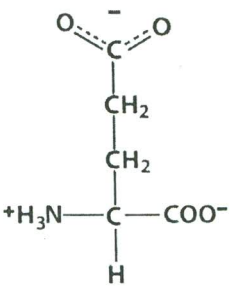
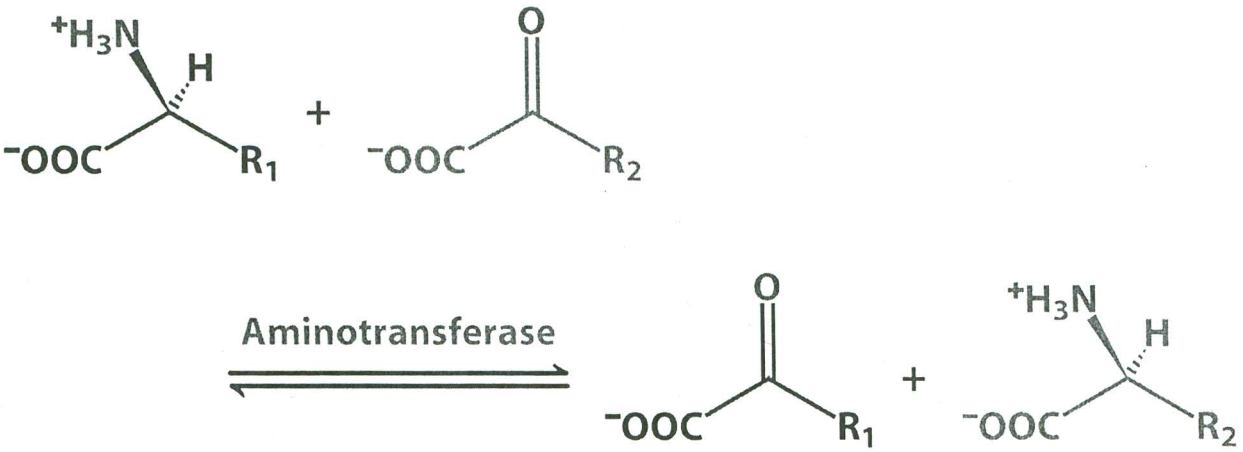

Glycolysis
Gluconeo genesis
ADP
ATP
CO_2 + ATP
Pi + ADP
GDP + CO_2
GTP
+ NH_4^+
NADH
NAD^+
Ala
Pyruvate
CoASH + NAD^+
pyruvate dehydrogenase
CO_2 + NADH
Acetyl-CoA
fatty acids
H_2O
CoASH
1. citrate synthase
Citrate
2. aconitase
cis-Aconitate
2. aconitase
Isocitrate
3. isocitrate dehydrogenase
NAD+
NADH + H+
Oxalosuccinate
3. isocitrate dehydrogenase
α-Ketoglutarate
CO_2
NADH + NH_4^+
NAD^+
Glu
4. α-ketoglutarate dehydrogenase
Succinyl-CoA
CoASH
NADH + H+
NAD^+
5. succinyl-CoA synthetase
GTP
GDP + P_i
Succinate
6. succinate dehydrogenase
FAD
$FADH_2$
Fumarate
7. fumarase
H_2O
L-Malate
8. malate dehydrogenase
NAD+
Oxaloacetate
NADH + NH_4^+
NAD^+
Asp
Citric acid cycle



Alanine (Ala, A)



Glutamate (Glu, E)



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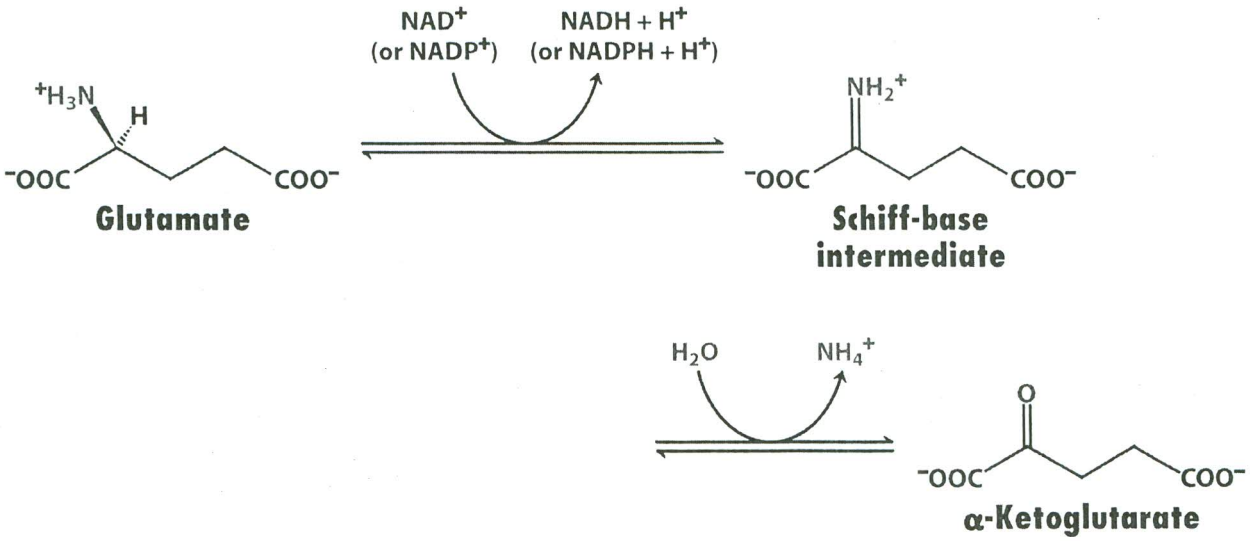

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